

# SANJANA SHAH

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## EDUCATION

### College of Natural Sciences and Mathematics, University of Houston

May 2025

Bachelor of Science in Computer Science, Minor in Mathematics

#### Relevant Coursework

Programming/Data Structures, Algorithms, Operating Systems, Computer Architecture, Databases, and Statistical Learning, Data Science, Software Engineering, Software Design, Artificial Intelligence, Statistical Learning, Analytical Learning, Business Principles

#### Awards

Dean's Scholars List - Fall 2023, Spring 2024, Fall 2024

## TECHNICAL SKILLS

**Programming Languages:** Python, SQL, R, C/C#/C++, Java, HTML, CSS, JavaScript, React JS, C#

**Databases:** MySQL, SQL server, MongoDB, PostgreSQL, NoSQL, AWS

**Machine Learning & Analytical Techniques:** Data Cleaning, Data Analysis, Data Visualization, Machine Learning Techniques, Statistical Analysis, Trend Analysis, Predictive Modeling

**Tools & Platforms:** Jupyter, VS Code, Git, GitHub

**Microsoft Tools:** Word, Excel, PowerPoint, Teams, Outlook, Microsoft Project

## EXPERIENCE

### Undergraduate Teaching Assistant | Department of Computer Science, University of Houston

January 2024 - Present

- Mentored 100+ students per semester in Python, SQL, and R, boosting class-wide performance on data analysis assignments by 25% through personalized instruction and group workshops.
- Conducted sessions on data preprocessing, statistical modeling, and visualization, resulting in a 40% increase in student confidence with data science tools and methods.
- Applied analytical thinking and instructional strategies to simplify complex data science topics, enhancing student engagement and retention of key concepts.

### Data Analyst Intern | Desi Brothers

September 2022 – April 2024

- Analyzed daily sales and inventory data to identify trends, reduce stockouts, and improve restocking efficiency by 20%.
- Created visual dashboards to track product performance and seasonal demand using charts and pivot tables.
- Streamlined manual data entry processes using Excel functions, reducing reporting time by 30%.
- Collaborated with store managers to provide actionable insights into pricing and promotions, supporting data-driven decision-making.

## PROJECTS

### The Buying Beacon: Customer Prediction | Python, SQL, XGBoost, Power BI, Pandas, E-commerce Analytics, ChatGPT, API

- Analyzed customer shopping behaviors using SQL for data extraction and Python (Pandas, Scikit-learn) for data preprocessing, feature engineering, and implementing XGBoost to predict purchase likelihood for new vs. returning customers.
- Visualized customer trends and insights using Power BI to create interactive dashboards, while employing Matplotlib and Seaborn for additional visual analytics, driving data-driven marketing strategies.

### NYC Airbnb Trends and Predictions | Python, MySQL, SQL, Tableau, Data visualization and preprocessing, Machine Learning

- Utilized Python (Pandas, NumPy) and MySQL to import, clean, and process large-scale Airbnb datasets, ensuring data accuracy and preparing it for analysis and machine learning applications. Built a price prediction model using XGBoost, evaluating performance with R-squared and MAE to provide actionable pricing insights.
- Developed interactive Tableau dashboards to visualize pricing trends, availability, and review patterns, enabling data-driven decision-making. Integrated data from MySQL and external sources to offer comprehensive insights on the NYC Airbnb market, facilitating easy exploration by stakeholders through visualizations and reporting in Tableau.

## CERTIFICATION

### Google Data Analytics Professional Certificate | Google | January 2025 | [Certifications](#)

- Completed an 8-course program with 180+ hours of hands-on training in data analytics, covering data cleaning, analysis, visualization, and R programming, and gained proficiency in SQL, Tableau, and spreadsheets
- Developed actionable insights through case studies and projects, mastering data preparation, statistical analysis, and visualization to communicate findings effectively to stakeholders
- Gained expertise in problem-solving, critical thinking, and data ethics for informed data analysis.